SecurePsych AI

Feature POC Checklist: RAG Service

Version 1.0 | CSIS494 | March 2026

Feature	RAG Q&A Service — Secure Document Question & Answer
Owners	Noble A., Mohammed A., Arham N.
Service Port	Microservice behind API gateway on port 8000
Framework	LangChain (confirmed) + ChromaDB (local embedded)
Purpose	Minimum bar for demo readiness and merge to main
Phase Gate	Phase 2: Core RAG + Summarization (Mar 11 – Mar 28)

1. What This Checklist Is

This is a PR acceptance checklist for the RAG service as a whole—not a task list. Before this service is considered demo-ready and eligible to merge to main, every item below must pass. Items are organized by concern: pipeline correctness, quality bar, and integration. Keep the bar practical: the goal is a working, trustworthy POC, not a production-hardened system.

2. Acceptance Criteria

#	Acceptance Criterion	Notes / Threshold
A. End-to-End Pipeline		
1	A user can upload a PDF document through the API and receive confirmation that ingestion completed successfully.	*Ingestion endpoint returns a success status with a document ID.*
2	The ingestion pipeline correctly extracts text from a multi-page PDF using PyMuPDF (fitz), chunks it with the parameters confirmed from Research 1 & 2, and stores embeddings in ChromaDB.	*Test on a 40–50 page document. Chunks must be stored without silent errors.*
3	A user can send a natural-language question and receive a grounded answer that references content from the uploaded document.	*Answer must be traceable to retrieved chunks; not fabricated.*
4	The response payload includes source citations with page numbers or section references for each retrieved chunk used to generate the answer.	*At least one citation per answer. Page reference format is flexible for POC.*
5	The default RAG mode (flat/semantic, ChromaDB) is active and operational as the standard query path.	*This is the path that must work reliably for demo.*
6	If hierarchical RAG mode is implemented: toggling 'extended thinking' on routes the query through the indexed RAG path and returns a noticeably different (richer) answer on a test document.	*If the toggle is not implemented by merge, mark*

		as deferred. Flat RAG alone is sufficient for Phase 2 merge.
B. Quality Bar - Ignore this for now since we don't have a machine.		
7	On a set of 5 representative test queries against a sample psychoeducational evaluation document, at least 4 out of 5 answers contain the correct factual information from the document.	*Informal pass/fail. Document which queries were tested.*
8	The service does not hallucinate facts that are absent from the uploaded document on any of the 5 test queries.	*Any hallucination is a blocker. The RAG architecture must ground answers in retrieved chunks only.*
9	Query latency for the default flat RAG mode is under 30 seconds on the target hardware for a 48-page document.	*Record actual measured latency in PR notes.*
10	LangSmith tracing is active on the RAG query chain and shows per-step latency and token counts. No document content is transmitted externally—only metadata.	*Verify with a network capture or LangSmith dashboard review.*
C. Gateway Integration - Ignore this for now since we have no frontend yet and tests are through cli		
11	The RAG service runs as an independent FastAPI microservice and registers its routes correctly through the API gateway on port 8000.	*The frontend never calls the RAG service directly.*
12	The /health endpoint for the RAG service returns a 200 OK response and is reachable through the gateway.	*Health check must pass before CI/CD marks the build green.*
13	The RAG service respects the feature flag in config.yaml. If the service is disabled via flag, the gateway returns a graceful 'feature unavailable' response rather than an error.	*Smoke test with the feature flag toggled off.*
14	Per-request temp files (chunked content, intermediate embeddings) are cleaned up after each session via the shared cleanup utility.	*Verify no orphaned files after 3 sequential test queries.*
D. Security & Compliance Baseline - Don't sweat this too much for now but security is the main thing here		
15	The service makes zero outbound network calls during a query. All LLM inference routes through the local Ollama instance.	*Confirm with: curl api.openai.com must fail (connection refused) while the service is running.*
16	ChromaDB data is stored within the shared encrypted storage path (LUKS/fscrypt). The service does not write student data outside this path.	*Spot-check file write locations during an ingestion run.*
17	All file writes must use the SECURE_STORAGE_PATH environment variable. No hardcoded paths.	*Verified by spot-checking during ingestion run.*

3. What Is Explicitly Out of Scope for This Checklist

The following are not required for Phase 2 merge and are tracked separately:

- Frontend UI for file upload and chat — tracked under Alyse B.'s frontend work.

- FERPA compliance validation — blocked pending Jon T.'s research completion.
- Hierarchical/indexed RAG mode — desirable but not a Phase 2 blocker. Flat RAG alone qualifies for merge.
- Performance under concurrent users — Phase 3 benchmark (target: 3 concurrent users).
- Multi-document / cumulative file cross-referencing — Phase 3+ scope.

4. How to Use This Checklist

When the RAG service is ready for merge, the owner (Noble, Mohammed, or Arham—coordinate internally) opens a PR and pastes this checklist into the PR description. Check off each item with evidence (e.g., test output, LangSmith screenshot, latency measurement). Natoli reviews the completed checklist before approving the merge. Any unchecked item needs an explicit decision: either fix it, defer it with justification, or accept it as a known gap.